

End-to-End ML Student Performance Predictor

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Machine Learning Project

Abstract—The End-to-End ML Student Performance Predictor is a machine learning-based web application developed to predict student academic performance using demographic and examination-related data. The project implements a complete machine learning workflow including data ingestion, preprocessing, model training, evaluation, model selection, and deployment through a Flask web application.

The system uses multiple regression algorithms such as Linear Regression, Decision Tree Regressor, Random Forest Regressor, Gradient Boosting Regressor, XGBoost, CatBoost, and AdaBoost to predict student math scores. The project also demonstrates preprocessing pipelines, hyperparameter tuning, model serialization, and real-time prediction systems.

The trained model is integrated with a Flask-based web application to provide real-time predictions through a user-friendly interface. The project follows a modular and scalable architecture suitable for production-style machine learning workflows.

Index Terms—Machine Learning, Student Performance Prediction, Regression Algorithms, Flask, Data Preprocessing, End-to-End ML Pipeline

I. INTRODUCTION

The End-to-End ML Student Performance Predictor is developed to predict student academic performance using machine learning techniques. The project demonstrates a complete machine learning lifecycle including preprocessing, training, evaluation, deployment, and inference.

The system predicts student math scores using features such as gender, race/ethnicity, parental education level, lunch type, test preparation status, reading score, and writing score.

The project aims to demonstrate production-level machine learning engineering concepts such as modular architecture, preprocessing pipelines, model deployment, and real-time inference.

A. Purpose

The main purpose of this project is to develop a complete end-to-end machine learning system capable of predicting student academic performance.

The project demonstrates:

- Data preprocessing pipelines
- Machine learning model training
- Hyperparameter tuning
- Model serialization
- Flask deployment
- Real-time prediction systems

B. Target Beneficiaries

The project is beneficial for:

- Students
- Educational Institutions

- Teachers and Mentors
- Researchers and Data Analysts
- Machine Learning Learners

C. Project Scope

The scope of the project includes:

- Data preprocessing and transformation
- Regression model training
- Hyperparameter tuning
- Model evaluation and selection
- Flask-based deployment
- Real-time prediction

II. PROJECT DESCRIPTION

The End-to-End ML Student Performance Predictor follows a modular architecture integrating preprocessing, model training, evaluation, and deployment.

The dataset is divided into training and testing datasets. Numerical and categorical features are transformed using preprocessing pipelines before training machine learning models.

The best-performing model is selected using the R^2 evaluation metric.

A. Algorithms Used

The following regression algorithms are used in the project:

- 1) Linear Regression
- 2) Decision Tree Regressor
- 3) Random Forest Regressor
- 4) Gradient Boosting Regressor
- 5) K-Nearest Neighbors Regressor
- 6) XGBoost Regressor
- 7) CatBoost Regressor
- 8) AdaBoost Regressor

B. Best Model Accuracy

Linear Regression achieved the highest R^2 Score of **88.04%**.

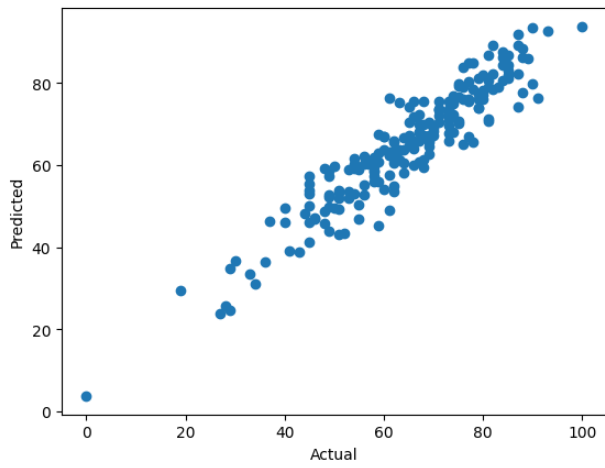


Fig. 1. Linear Regression Highest Accuracy

D. Workflow Architecture

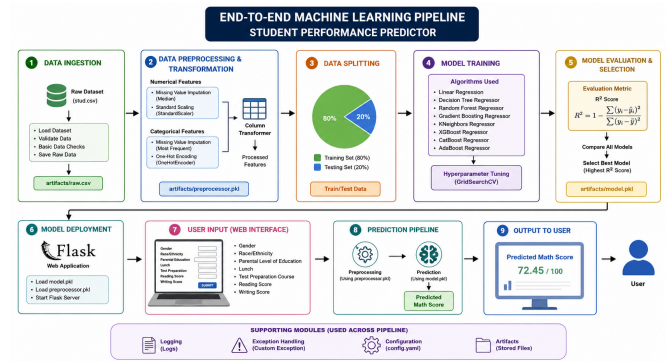
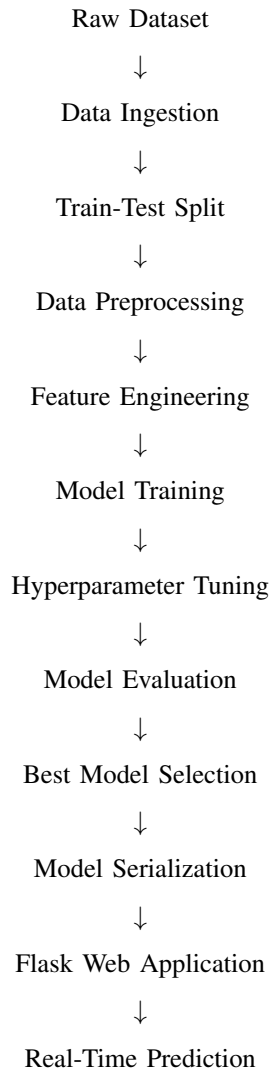


Fig. 2. End-to-End ML Workflow Architecture

C. Project Pipeline



E. Preprocessing Techniques

Data preprocessing converts raw data into structured data suitable for machine learning models.

1) Numerical Preprocessing:

- Missing Value Imputation
- Feature Scaling using StandardScaler

Standardization Formula:

$$z = \frac{x - \mu}{\sigma}$$

2) Categorical Preprocessing:

- Missing Value Handling
- One-Hot Encoding

III. GRAPHS AND PLOTS

A. Histogram

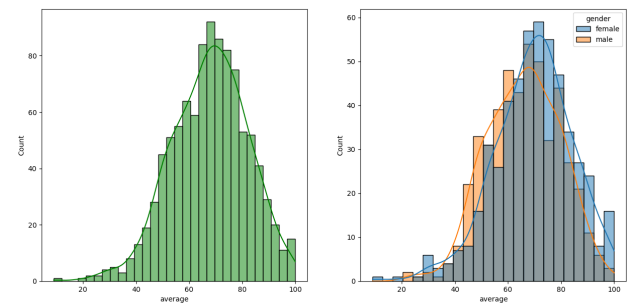


Fig. 3. Histogram Plot

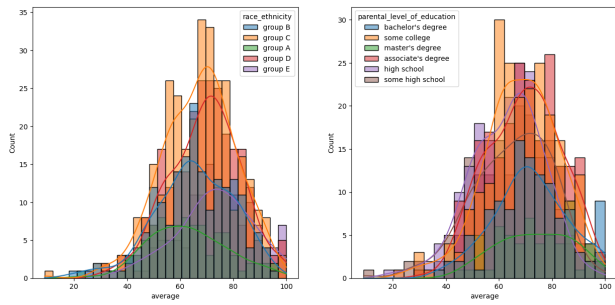


Fig. 4. Histogram Plot

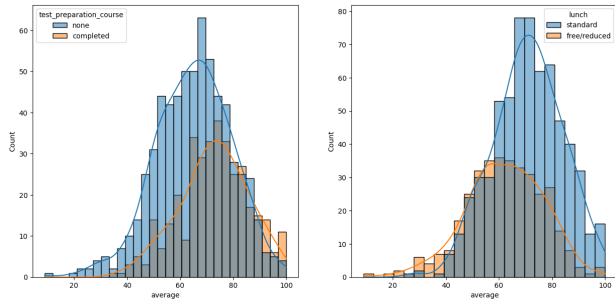


Fig. 5. Histogram Plot

D. Violin Plot

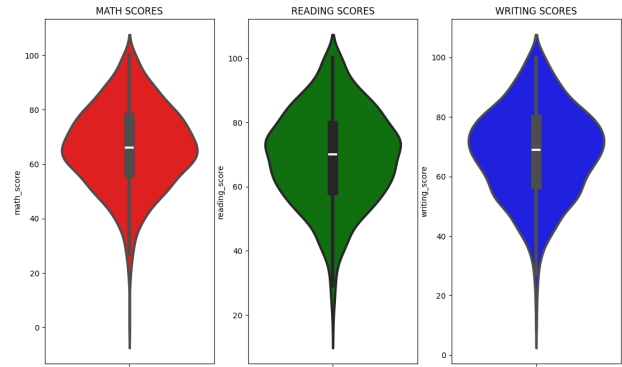


Fig. 8. Violin Plot

B. Box Plot

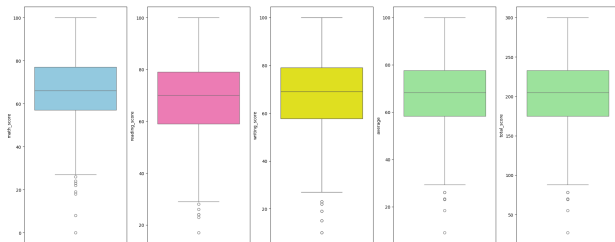


Fig. 6. Box Plot

C. Pie Plot



Fig. 7. Pie Plot

	Model_Name	R2_Score
2	Ridge	0.880593
0	Linear Regression	0.880433
10	AdaBoostRegressor	0.853578
8	CatBoost	0.851632
7	Random Forest Regressor	0.849360
9	XGBRegressor	0.827797
1	Lasso	0.825320
4	K-neighbour Regressor	0.783813
6	Decision Tree Regressor	0.743834
3	Elastic Net	0.739624
5	SV Regressor	0.728600

Fig. 9. Comparative analysis

IV. DEPENDENCIES

TABLE I
SOFTWARE DEPENDENCIES

Dependency	Purpose
Python	Python 3.8
Flask	Web Framework
Scikit-learn	Machine Learning
Pandas	Data Manipulation
NumPy	Numerical Computation
XGBoost	Gradient Boosting
CatBoost	Categorical Boosting
Matplotlib	Visualization
Seaborn	Statistical Plotting

V. CONCLUSION

The End-to-End ML Student Performance Predictor successfully demonstrates the implementation of a complete machine learning workflow for predicting student academic performance.

The project integrates data preprocessing, model training, evaluation, deployment, and real-time prediction into a unified architecture.

The Flask deployment provides a user-friendly interface for prediction, while the modular architecture ensures scalability and maintainability.

REFERENCES

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